THE MAXIMUM STRATUM OF THE GEOMETRIC FIBRE COUNT IS OPEN WITH FINITE PREIMAGE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 22030-22062, immediately following the statement at 22018-22028.

Editorial note. One 'some details omitted' phrase at source line 22045 (the identification d = n = sum_i m_i in the splitting) is retained verbatim and flagged for the reviewer by the omitted-steps note.

Editorial note. Two same-chapter pointers (count-geometric-fibres family, splitting lemma) and one cross-chapter pointer (Descent) are recorded in assembly.external_references.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 22048 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Upper semicontinuity of the fibre count and openness of its maximum. Lemmas lemma-count-geometric-fibres-universally-open (cited at proof lines 22032-22033) and lemma-count-geometric-fibres of this chapter. Supplies openness of Y_d = {n = d} directly, and defines the function n_{X/Y} whose base-change compatibility is used later in the proof.
- Etale splitting off the quasi-finite part, technical variant. Lemma lemma-etale-splits-off-quasi-finite-part-technical-variant of this chapter, cited at proof lines 22037-22039. Provides, around each point y, an etale neighbourhood U -> Y with U x_Y X = W disjoint-union V_{i,j}, finitely many etale pieces V_{i,j} over U plus a remainder W.
- Descending the finite locus property along an open cover. Descent, Lemma descent-lemma-descending-property-finite, cited at proof lines 22057-22058. Finiteness of U x_Y X -> U over the etale neighbourhood implies finiteness of X -> Y over the image of U -> Y, which is open because f is universally open.

Source reference. Original reference 02LA: focused-seeds/geometry/S9-morph-large-open/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0F35: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 21960–21960. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0F36: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 21981–21981. Navigation pointer only: the counted

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proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 02LO: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12136–12136. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ be a separated, locally quasi-finite, and universally open morphism of schemes. Let $n_{X/Y}$ be as in Lemma 0F35. If $n_{X/Y}$ attains a maximum $d < \infty$, then the set*

$$Y_d = \{y \in Y \mid n_{X/Y}(y) = d\}$$

is open in Y and the morphism $f^{-1}(Y_d) \to Y_d$ is finite.

Proof. The openness of Y_d is immediate from Lemma 0F36. To prove finiteness over Y_d we redo the argument of the proof of that lemma. Namely, let $y \in Y_d$. Then there are at most d points of X lying over y. Say $x_1, \ldots, x_n$ are the points of X lying over y. Apply Lemma 02LO to get an étale neighbourhood $(U, u) \to (Y, y)$ and a decomposition

$$U \times_Y X = W \amalg \coprod\nolimits_{i=1,\ldots,n} \coprod\nolimits_{j=1,\ldots,m_i} V_{i,j}$$

as in locus citatus. Observe that $d = n_{X/Y}(y) = \sum_i m_i$ in this situation; some details omitted. Since f is universally open, we see that $V_{i,j} \to U$ is open for all i, j. Hence after shrinking U we may assume $V_{i,j} \to U$ is surjective for all i, j and we may assume U maps into W. This proves that $n_{U \times_Y X/U} \geq \sum_i m_i = d$. Since the construction of $n_{X/Y}$ is compatible with base change we know that $n_{U \times_Y X/U} = d$. This means that W has to be empty and we conclude that $U \times_Y X \to U$ is finite. By Descent, Lemma 02LA this implies that $X \to Y$ is finite over the image of the open morphism $U \to Y$. In other words, we see that f is finite over an open neighbourhood of y as desired. □